Sarthak Mistry

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SUMMARY

Data Engineer with 2+ years building production batch and real-time streaming pipelines (Kafka, Debezium CDC, Spark, Airflow, Snowflake) at financial-services scale: 5M+ daily records in batch, 200K+ updates/hour in streaming. Experienced in dimensional and Data Vault modeling, data quality frameworks (Great Expectations, dbt tests), and cloud infrastructure.

PROFESSIONAL EXPERIENCE

Indiana University, Bloomington — ***Bloomington, IN***

Graduate Research Assistant — Data Infrastructure — ***Jan 2026 – May 2026***

- Integrated multi-cloud storage backends (AWS S3, GCS, Azure Blob) into the R interface of OpenBT, an open-source Bayesian ML library, enabling portable model persistence across cloud environments.
- Containerized the full analysis environment (Docker, RStudio) for reproducible cross-platform deployment, and designed validation benchmarks for the library's C++ to Rust migration.

Piramal Capital & Housing Finance — ***Mumbai, India***

Data Engineer — ***Jul 2022 – Jul 2024***

- Built batch ETL pipelines (Spark, Airflow) ingesting 5M+ daily records from Salesforce, MongoDB, SQL Server, and REST APIs into Snowflake, powering analytics across the lending business.
- Deployed real-time CDC pipeline (Debezium, Kafka, Spark Structured Streaming) processing 200K+ updates/hour at sub-minute latency, feeding live operational dashboards (Grafana).
- Implemented a 150-feature regulatory data mart (AWS Lambda, S3, Snowflake), cutting the compliance team's audit-reporting turnaround by ~40%.
- Built automated PII detection-and-masking pipeline (regex classification, Snowflake dynamic data masking) achieving 93% precision across 1M+ customer records.
- Optimized AWS Lambda flat file to Snowflake ingestion, increasing throughput by 46% while maintaining daily refresh SLAs.
- Partnered with the Chief Economist to automate an economic-indicator scraping pipeline (Python, Selenium), reducing manual effort by 73%.

NSEIT — ***Mumbai, India***

Data Engineering Intern — ***Oct 2021 – Mar 2022***

- Engineered CI/CD pipeline for 20+ dbt models on PostgreSQL, automating data modeling and reducing dashboard query latency by 40%.
- Designed automated dbt schema validations and data quality tests, reducing ingestion errors by 25% for Kibana visualizations.

PORTFOLIO PROJECTS

E-commerce Data Pipeline — *Airflow, dbt, Snowflake, Great Expectations* — GitHub

- Built production-grade medallion pipeline (RAW → staging → Data Vault 2.0 → star schema/OBT) across 17 dbt models on TPC-DS SF10 (~10GB, 25 tables) in Snowflake.
- Orchestrated 2 Airflow DAGs (Astro CLI, Astronomer Cosmos) with per-model retries; enforced 50+ Great Expectations checks across 7 suites with quarantine routing for analyst triage instead of hard pipeline failure.
- Modeled Data Vault 2.0 with MD5 hash-diff and SCD Type 2 history; built OBT with RFM customer segmentation.

Real-Time Transaction Streaming & Anomaly Detection — *Kafka, Spark, Hudi, Trino* — GitHub

- Designed self-managed 3-node Kappa-architecture cluster (Kafka, Spark, Trino) processing ~500 live Bitcoin transactions/min from a WebSocket feed.
- Deployed SparkML KMeans anomaly-detection model as a Spark Structured Streaming job scoring transactions in-flight, persisting results to Apache Hudi tables on HDFS.
- Exposed scored results through Trino with Hive Metastore for low-latency ad-hoc SQL investigation of flagged transactions.

SnowFlow Lineage — *React, Vis.js, Django, Snowflake* — GitHub

- Built full-stack lineage visualization tool (React/Vis.js, Django API) mapping Snowflake table and column dependencies from account metadata.
- Cut column-level impact analysis from ~30 minutes of manual SQL tracing to under 10 seconds of visual exploration.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, JavaScript
Data Engineering: Snowflake, dbt, Apache Airflow, Kafka, Spark (PySpark, Structured Streaming), Debezium CDC, Trino/Presto, Apache Hudi, ELT/ETL
Data Modeling & Quality: Star Schema, Data Vault 2.0, SCD Type 2, Kimball Dimensional Modeling, OBT, Great Expectations
Cloud & Infrastructure: AWS (S3, Lambda, EMR), GCP (BigQuery), Docker, Git, CI/CD
Governance & BI: Snowflake Dynamic Data Masking, RBAC, PII Detection, Grafana, Power BI, Streamlit

EDUCATION

University of Mumbai — **Aug 2018 – May 2022**

Bachelor of Engineering in Electronics and Telecommunication Engineering — GPA: 3.6

Indiana University, Luddy School of Informatics, Computing and Engineering — **Aug 2024 – May 2026**

Master of Science in Data Science — GPA: 3.6